

```
CREATE EXTENSION hstore;
```

Create the table with the *hstore* type to store the features of a smartphone:

```
CREATE TABLE smartphones (
  id serial PRIMARY KEY,
  name VARCHAR(255),
  features hstore);
```

Set the contents of the *hstore* as key-value pairs:

```
INSERT INTO smartphones (name, features) VALUES (
  'Asus Zenfone 6',
  'os      => "Android v4.3 (Jelly Bean)"
  chipset => "Intel Atom Z2580",
  cpu     => "Dual-core 2 GHz",
  gpu     => "PowerVR SGX544MP2"
);
```

An SQL query to retrieve data from an *hstore*:

```
SELECT name, features->'os' AS os
FROM   smartphones
WHERE  features @> 'cpu' => "Dual-core 2 GHz" ::hstore
```

With the current 9.4/9.5 PostgreSQL release, *GIN* and *GIST* indexes now support *hstore* columns with excellent performance metrics.

NewSQL

With the Infrastructure as a Service (IaaS) model, the biggest shift in architecture design is that consistency, transactions and storage optimisation are all sought in the cloud as we seek scalability at all levels.

NewSQL is an emergent class of database systems that have a combination of RDBMS and NoSQL capabilities. NewSQL seeks to provide the scalability of NoSQL data stores for online transaction processing (OLTP) read-write workloads, while maintaining the ACID (atomicity, consistency, isolation and durability) property of RDBMS. Examples are NuoDB (a distributed cloud based database), VoltDB (high performance through in-memory operations), ScaleDB and GenieDB (both plug into MySQL), and Clustrix, which can work with any SQL database.

Figure 6 lays out the comparison between the various approaches.

Re-CAP Brewer's theorem

According to the CAP theorem, any networked, shared-data system can have at most two of three desirable properties, that is, (C)onsistency, (A)vailability and (P)artition.

Without having a clear understanding of what the CAP theorem states or rather what it restricts, most geek discussions project it as a '2 of 3' formulation. This is not

helpful! The CAP theorem helps explore design space, when we seek an answer to the following questions: "What capabilities do we seek in a system during a network partition?" and "What expectation can we have about the state of the system during partition recovery."

As Brewer himself observes:

- The state on both sides must become consistent
- There must be compensation for the mistakes made during the partition mode

Version vectors and commutative replicated data types (CRDT) are some of the newer approaches to manage invariants during partitions.

Thus, the choice of a data-store will depend heavily on the service's invariants and operations. 

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